Figure 1: gPXE connection timed out failure

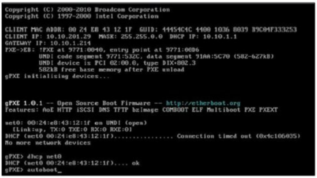

Figure 2: gPXE shell howto

steps are as below:

a. Download the latest gPXE source (1.0.1) tar ball.

b. Uncompress the source as shown below:

```
~# tar xvfz gpxe-1.0.1.tar.gz #Assuming gpxe-
1.0.1.tar.gz is the downloaded filename.
~# cd gpxe-1.0.1/
```

c. Change *'#undef TIME_CMD'* to *'#define TIME_CMD'* in *src/config/general.h*

d. Create a custom script (say *sleep.gpxe*) in the *src* directory with the following content:

```
#!gpxe
echo "Greetings... Running through the custom
script..."
 sleep 10            # For some NICs, a sleep
of 5 seconds may be good enough. We tested it
on
couple of Broadcom NICs which required a 10
sec. delay to get an IP
echo "Fetching DHCP IP for the network
adapter"
ifopen net0
dhcp net0
autoboot
```

e. Recompile the source to create a custom gPXE image with the script included as follows:

```
~# make clean
~# make bin/undionly.kpxe EMBEDDED_
IMAGE=sleep.gpxe
```

Copy the *undionly.kpxe* created under *bin* to the TFTP server.

Using the gPXE shell: This workaround is to pass the commands via the gPXE shell to fetch the DHCP IP and boot into the gPXE menu. Press *CTRL-B* when the screen prompts you right after the connection timeout. That brings

Figure 3: gPXE continues to boot with workaround

you the gPXE> shell shown in Figure 2.

Using the gPXE shell prompt—the second option: This workaround is again based on the gPXE shell. If there is no *menu.cfg* created in your Web server, you may need to manually enter the OS details via the shell as shown below:

```
gPXE>dhcp net0
gPXE>kernel -n mboot.c32 http://< WebserverIP >/
mboot.c32
gPXE>imgargs mboot.c32 -c http://< WebserverIP >/
boot.cfg
gPXE>boot mboot.c32
```

Note that the above commands are specific to VMware ESXi 5.x. It may be slightly different for Linux.

Test your gPXE set-up

If you have chosen the first workaround, you may see the result shown in Figure 3.

This article addresses the specific timeout issue that we encounter while using gPXE. This is useful for systems administrators who plan to deploy various operating systems over the network using gPXE/iPXE. END

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